

Sub-geometric Convergence of MALA

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Convergence rates of the Metropolis-adjusted Langevin algorithm (MALA) have not been well-studied in the regime that the target distribution π has a density with lighter than Gaussian tails. It is well-known that MALA is not geometrically ergodic in this setting, but sub-geometric rates have not been established. This contrasts with the heavy-tailed setting, in which polynomial convergence rates have been established in [JR07]. We consider a model problem setup to do this.

1 Setup

Consider the target distribution with density $\pi(x) \propto e^{-U_p(x)}$. The MALA proposal takes the form

$$y = x - \frac{h^2}{2} \nabla U(x) + h\xi,$$

where $\xi \sim N(0, I_{d \times d})$ and d is the dimension. The acceptance rate is

$$\alpha(x, y) = 1 \wedge \exp(-\Delta(x, y)),$$

where

$$\Delta(x, y) = U(y) - U(x) + \frac{1}{2} (x - y) \cdot (\nabla U(x) + \nabla U(y)) + \frac{h^2}{8} (\|\nabla U(y)\|^2 - \|\nabla U(x)\|^2).$$

We can consider the model

$$U_p(x) = \|x\|^{2p}, \quad \nabla U(x) = x\|x\|^{2p-2},$$

where $\|\cdot\|$ denotes the Euclidean norm and $p > 1$ is a positive integer. In this setting

$$\lim_{\|x\| \rightarrow \infty} \frac{\|\nabla U(x)\|}{\|x\|} = \infty,$$

from which it follows that MALA is not geometrically ergodic [RT96].

To establish sub-geometric convergence we must find a function $V : \mathbb{R}^d \rightarrow [1, \infty)$, which is usually monotonically increasing, $\rightarrow \infty$ as $\|x\| \rightarrow \infty$ and for which $\int V(x)\pi(dx) < \infty$, such that

$$PV(x) - V(x) \leq -\phi \circ V(x) + b\mathbb{I}_C(x),$$

where $PV(x) := \int V(y)P(x, dy)$, $b < \infty$, C is a ‘petite’ set (which typically means a compact set) and ϕ is some concave monotonically increasing function that $\rightarrow \infty$ as its argument does [DFMS04]. Typical choices for $V(x)$ are $\log(c + \|x\|)$, $1 + \|x\|^b$ or $e^{s\|x\|}$, but it can also be chosen to depend more directly on $U(x)$. $\phi(t) = t^{1-\delta}$ for some $\delta < 1$ or $\phi(t) = \log(t)$ are common choices of the monotonic function ϕ .

2 Model problem

Consider first the one-dimensional set up. The key idea is to split the integral

$$\int [V(y) - V(x)]P(x, dy) = \int [V(y) - V(x)]\alpha(x, y)Q(x, dy)$$

into distinct regions $A_1, \dots, A_p$. In some regions the integrand is negative, in others it is positive. The game is to show that overall the integral is negative, and sufficiently so to counteract $\phi \circ V(x)$ as $\|x\| \rightarrow \infty$. This will essentially prove the result.

To do this we will need to counterbalance the rate at which V grows with the rate at which $\alpha(x, y)$ decays when $\|y\|$ is large and $Q(x, A_1)$ decays when $A_1 := \{\|y\| < \|x\| - \epsilon\}$ or $\{\|y\| \leq (1 - \epsilon)\|x\|\}$, or some other suitable set in which the integral is clearly large and negative. In this regime both $Q(x, A_1)$ and $\alpha(x, y)$ when $\|y\| \gg \|x\|$ will decay to zero rapidly, and these things must be carefully bounded above/below to control the behaviour of the integral.

I am not sure if it is possible to establish such a sub-geometric bound, but beginning on the model problem and starting in one-dimension, numerical integrations can be performed for various choices of V and increasing sequences of $|x|$ to better understand the problem if needs be.

References

- [DFMS04] Randal Douc, Gersende Fort, Eric Moulines, and Philippe Soulier. Practical drift conditions for subgeometric rates of convergence. *Ann. Appl. Probab.*, 14(1):1353–1377, 2004.
- [JR07] Søren F Jarner and Gareth O Roberts. Convergence of heavy-tailed monte carlo markov chain algorithms. *Scandinavian Journal of Statistics*, 34(4):781–815, 2007.
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